

Digital Forensic Analysis of a Phishing Email Impersonating DHL

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Environment: Windows OS, Chrome Browser & Roundcube Webmail

Project link: <https://github.com/OluwaseunOa/Cybersecurity-Projects/tree/main/Cyber-Security-Fundamentals>

This report documents the steps taken during the analysis of a suspicious email claiming to be from DHL. The aim is to determine whether the message was legitimate or a phishing attempt, and to document the investigative process for awareness and training.

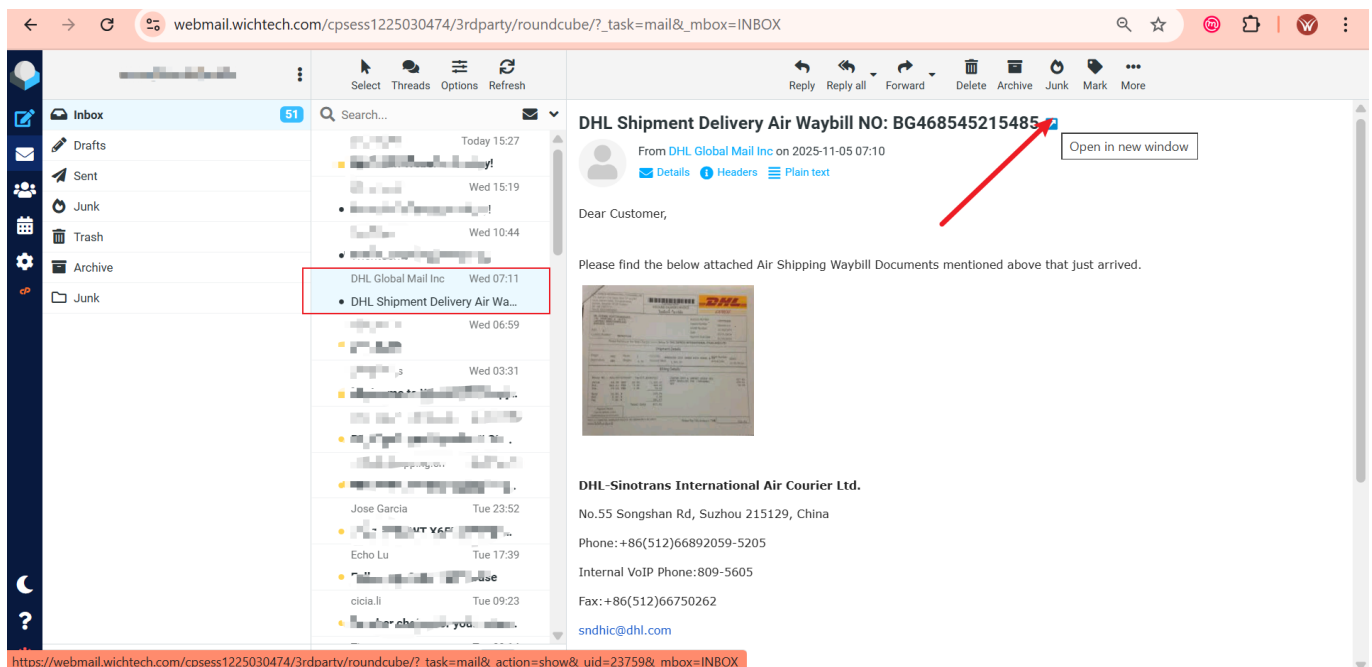
Project Overview

A suspicious email was received with a subject line suggesting an urgent DHL shipment delivery notice. The visual appearance of the email attempted to create a sense of urgency to prompt the recipient to click a link or open an attachment.

This report walks through each step used to verify the authenticity of the email.

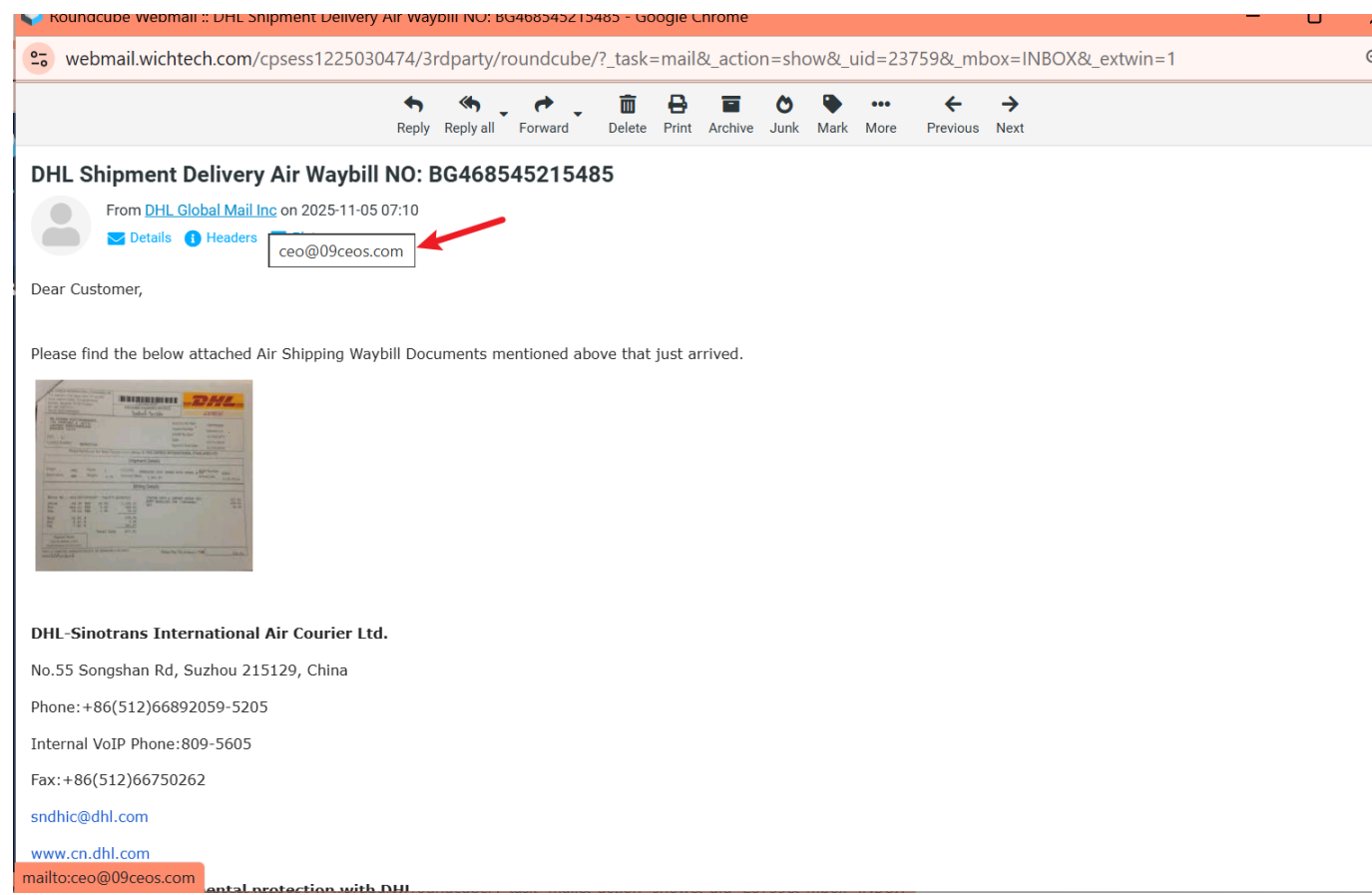
Investigation Steps

1. Received Suspicious Email



The email subject appeared to reference a DHL shipment, which is a common tactic used in phishing to create urgency.

2. Checked Sender Address



Hovering over the sender name revealed the email domain did not belong to DHL.

3. Opened Message Headers

Roundcube Webmail :: DHL Shipment Delivery Air Waybill NO: BG468545215485 - Google Chrome

webmail.wichtech.com/cpsess1225030474/3rdparty/roundcube/?_task=mail&_action=show&_uid=23759...

ReplyReply allForwardDeletePrintArchiveJunkMarkMorePreviousNext

DHL Shipment Delivery Air Waybill NO: BG468545215485

From DHL Global Mail Inc on 2025-11-05 07:10

Details

Headers

Plain text

Dear Customer,

Please find the below attached Air Shipping Waybill Documents mentioned above that just arrived.



DHL-Sinotrans International Air Courier Ltd.

No.55 Songshan Rd, Suzhou 215129, China

Phone: +86(512)66892059-5205

Internal VoIP Phone: 809-5605

Fax: +86(512)66750262

sndhic@dhl.com

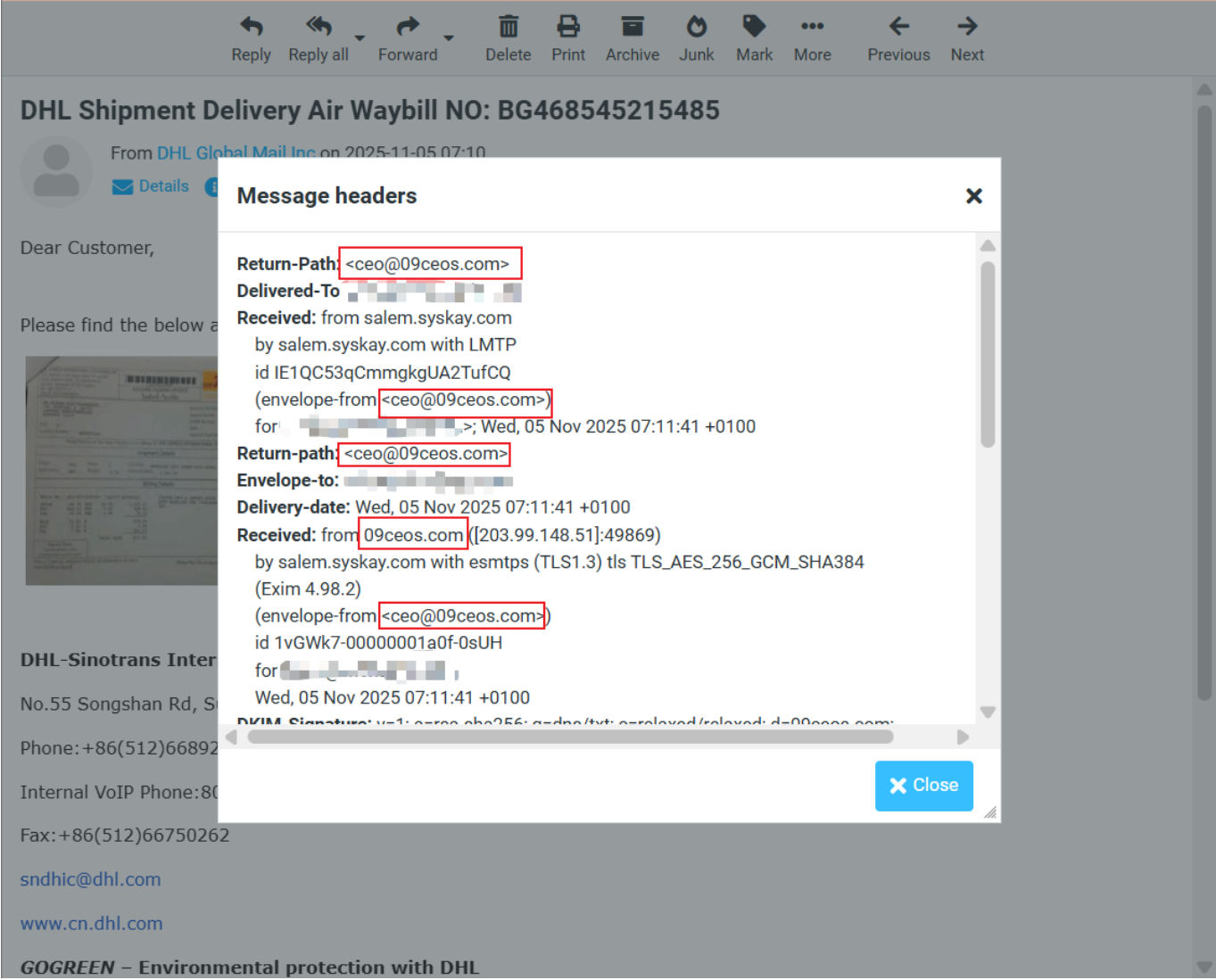
www.cn.dhl.com

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The headers were opened to verify details about the sending server and domain authentication.

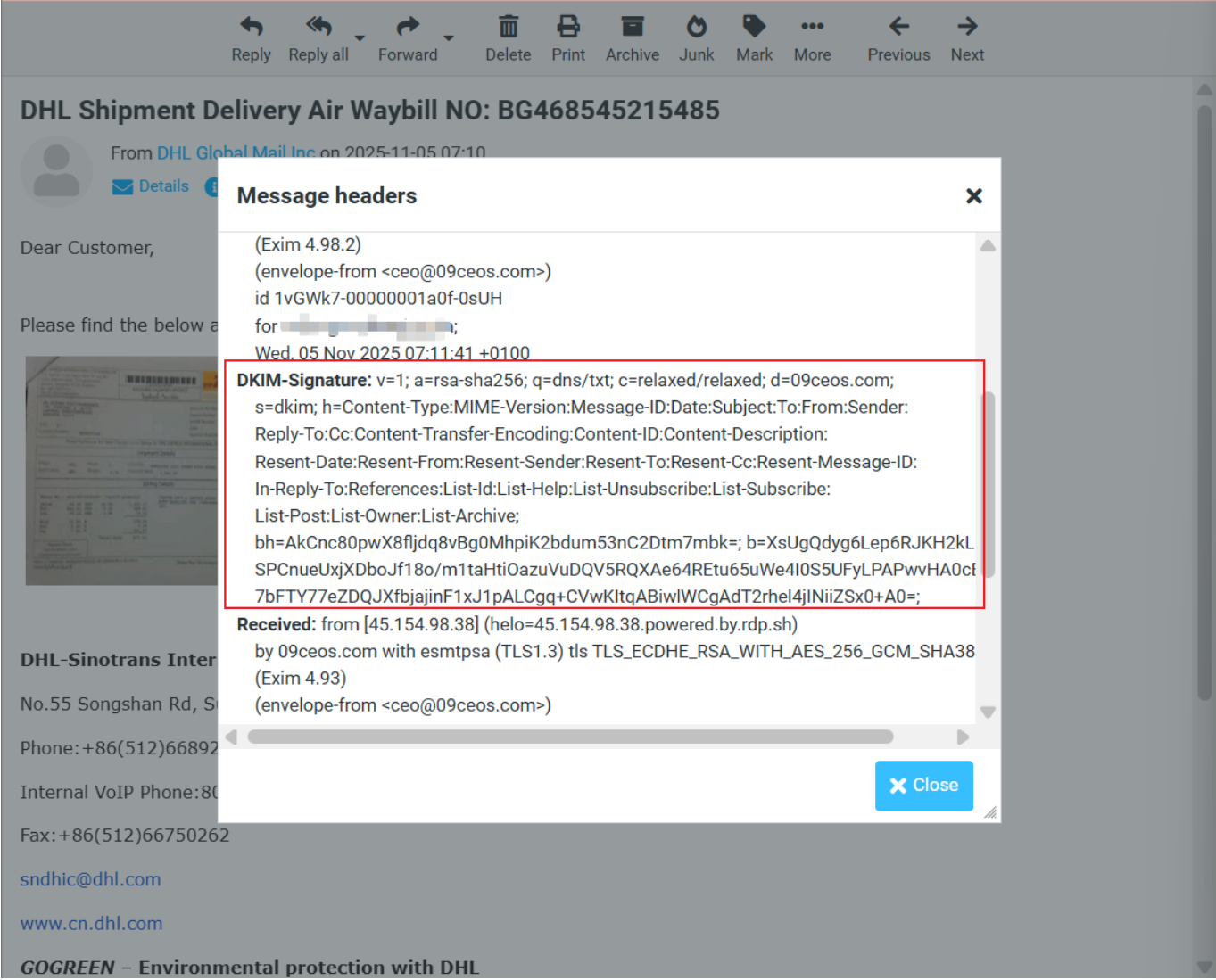
4. Sender Domain Mismatch

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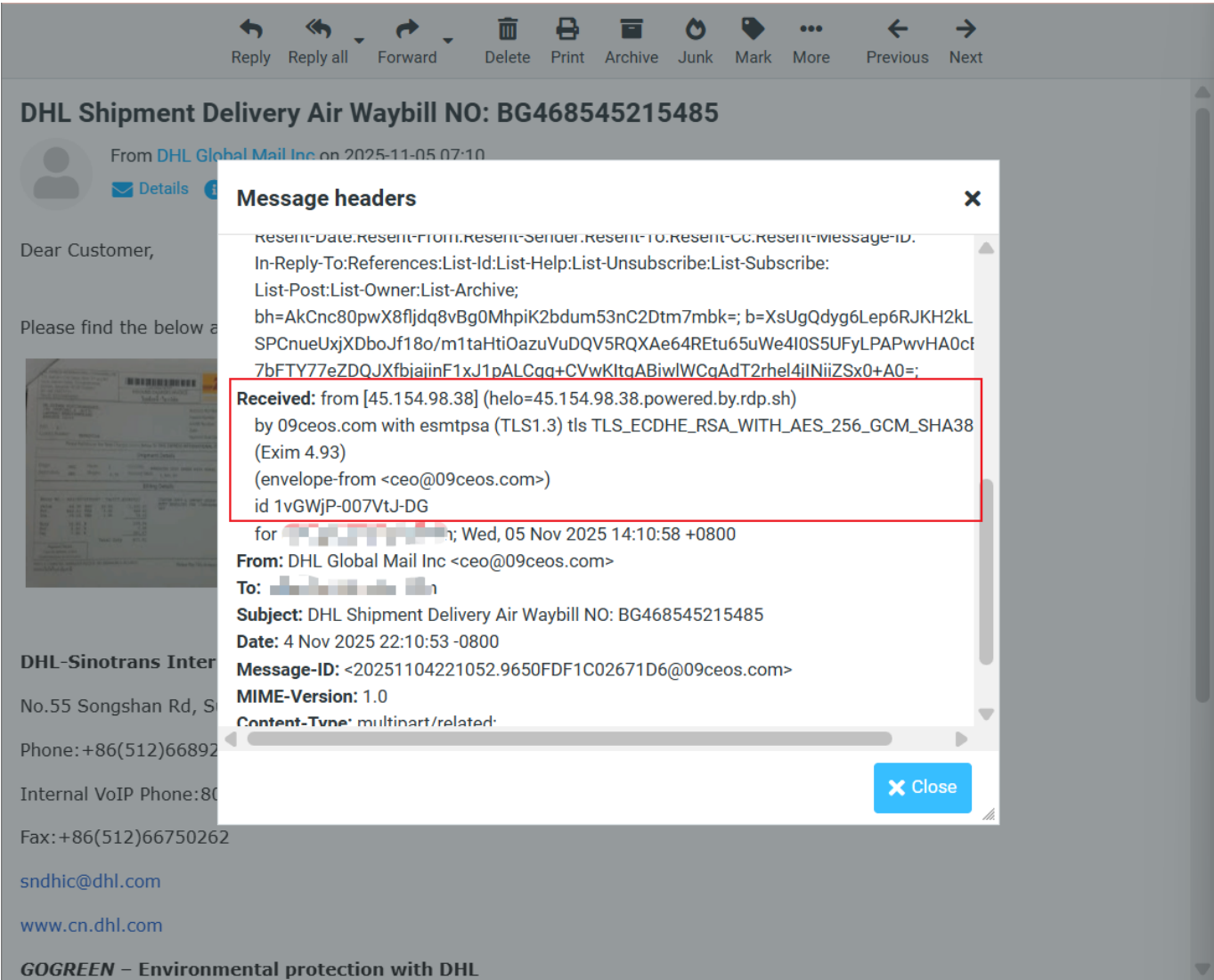
The "From" address did not match official DHL systems.

5. DKIM Signature Review



The DKIM signature only verified the domain 09ceos.com, not DHL.

6. Suspicious Mail Routing

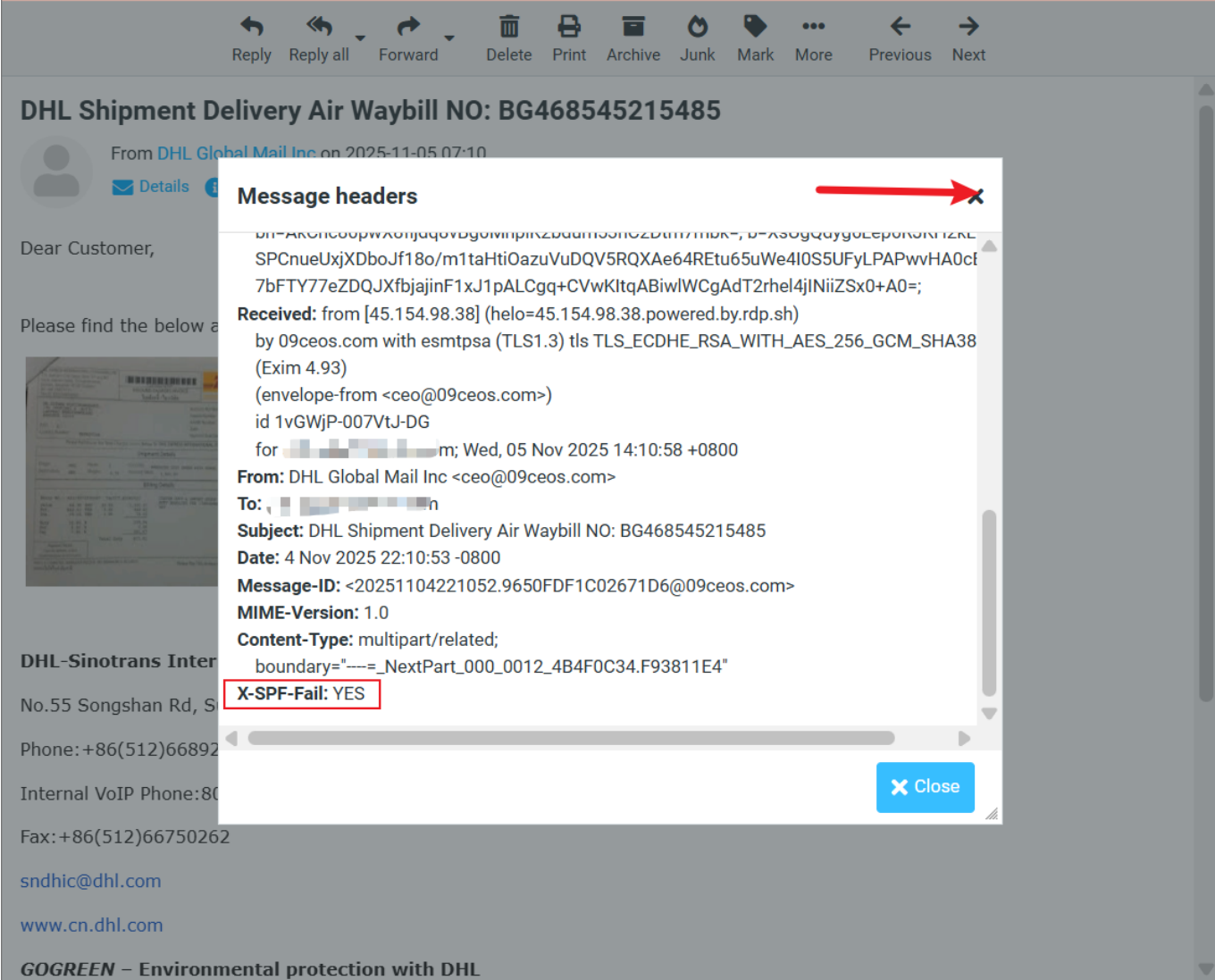


Routing traced through a server associated with compromised RDP hosts.

7. Urgent Subject Pattern

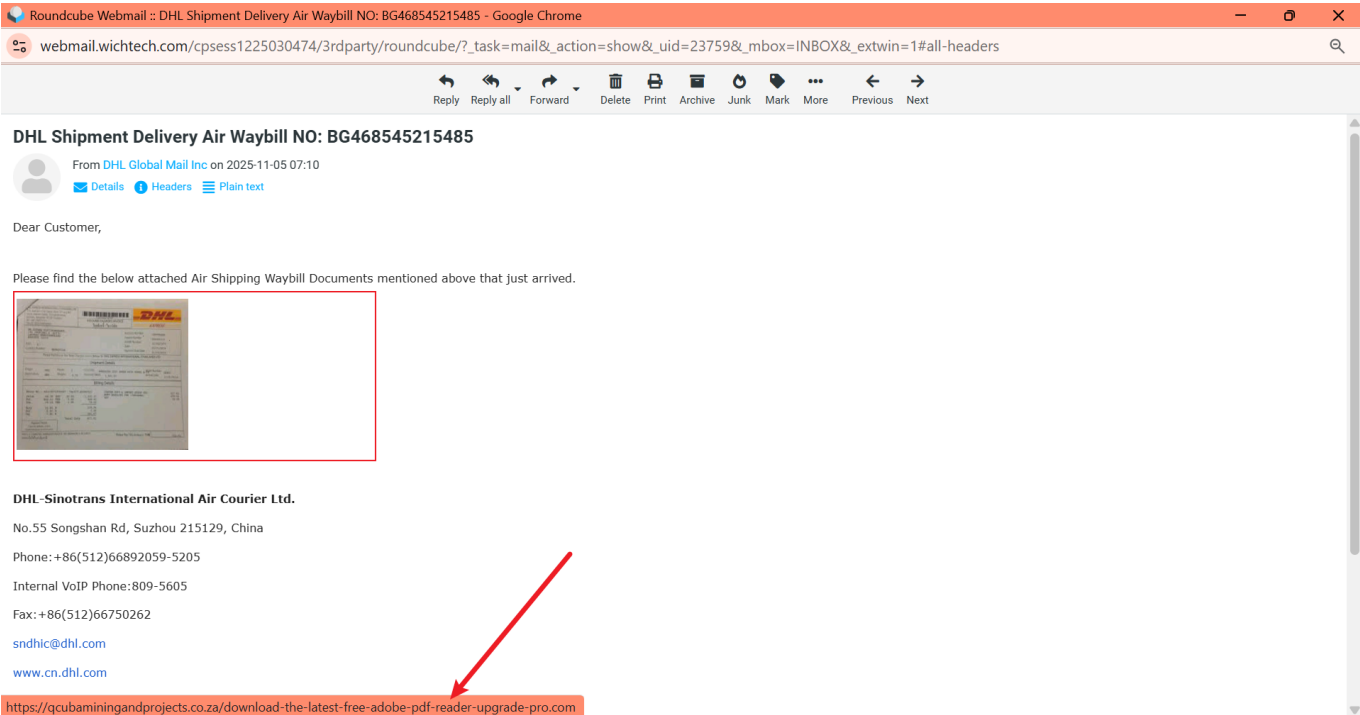
 Urgency Pattern The subject mimicked DHL/FedEx shipment notices, a known phishing tactic.

8. SPF Failure Confirmed



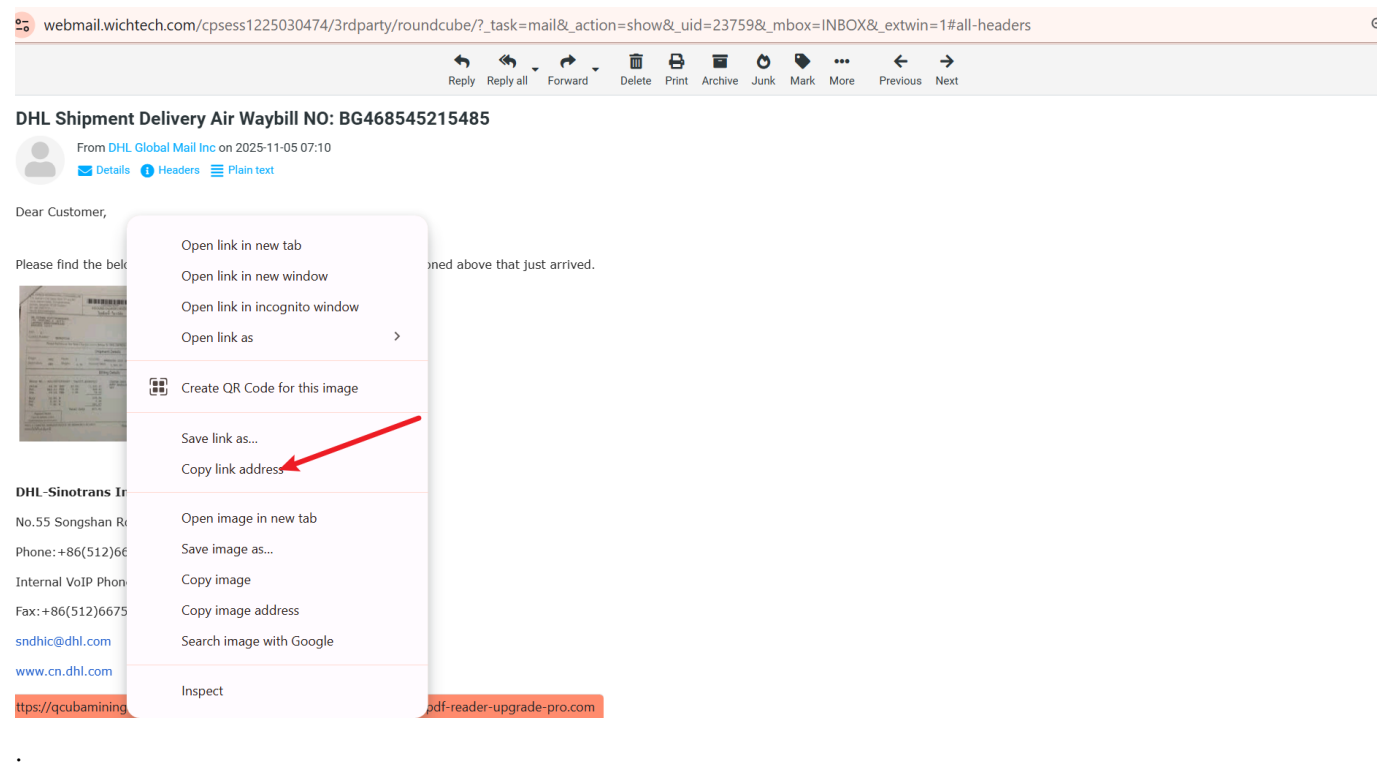
The email failed SPF validation, meaning the sender was not authorized to send from that domain.

9. Suspicious Embedded Image Link

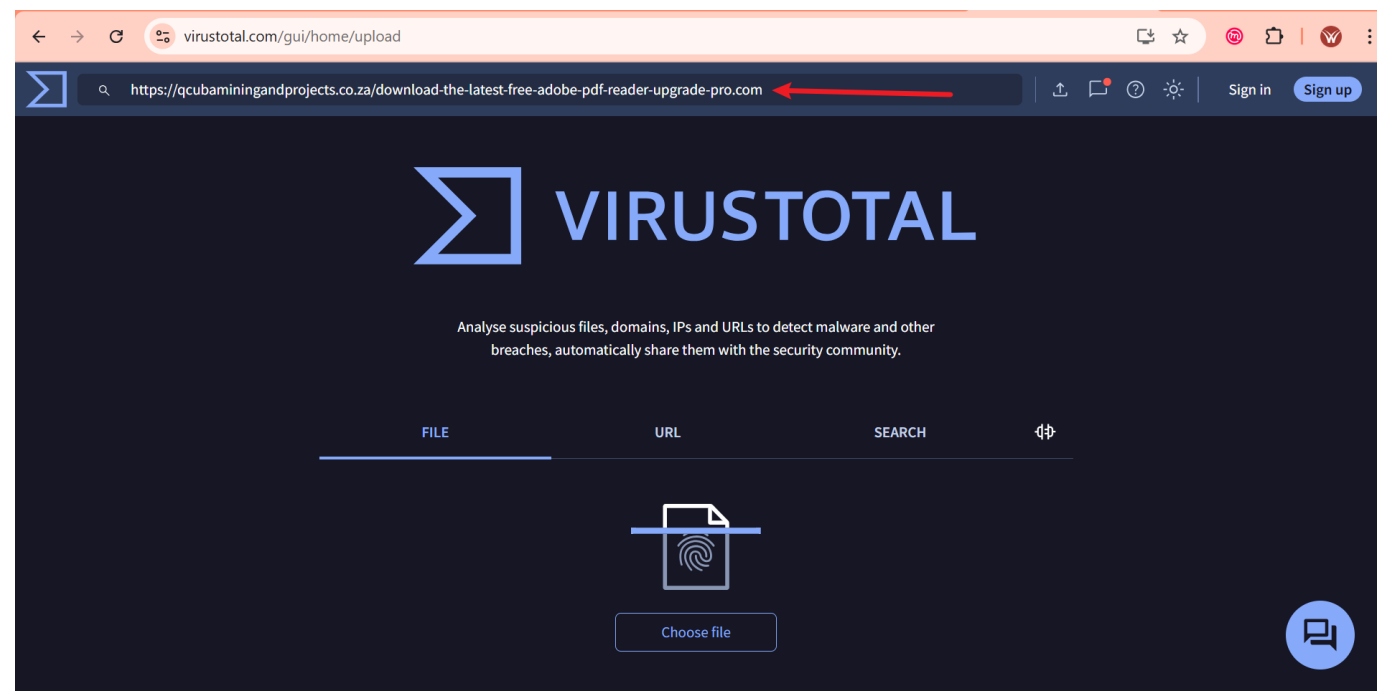


The embedded "document" image contained a dangerous external link.

10. Extracted Link for Analysis

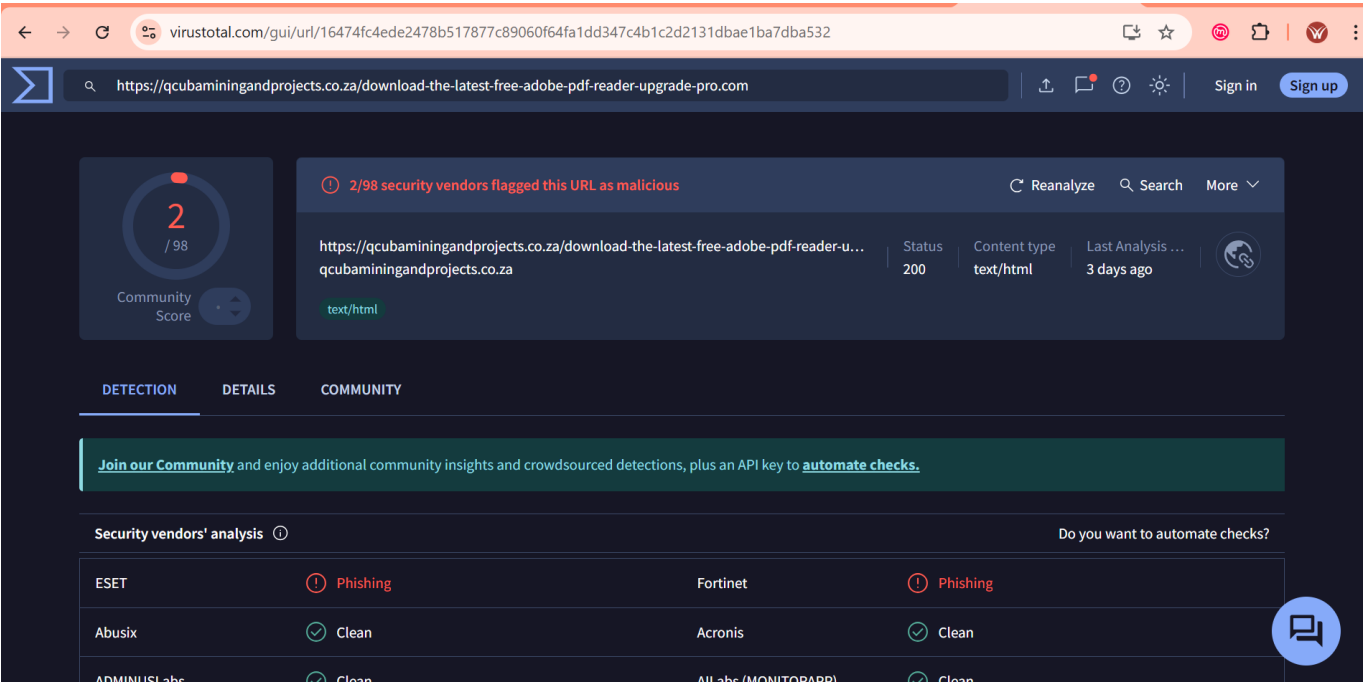


11. Submitted to VirusTotal



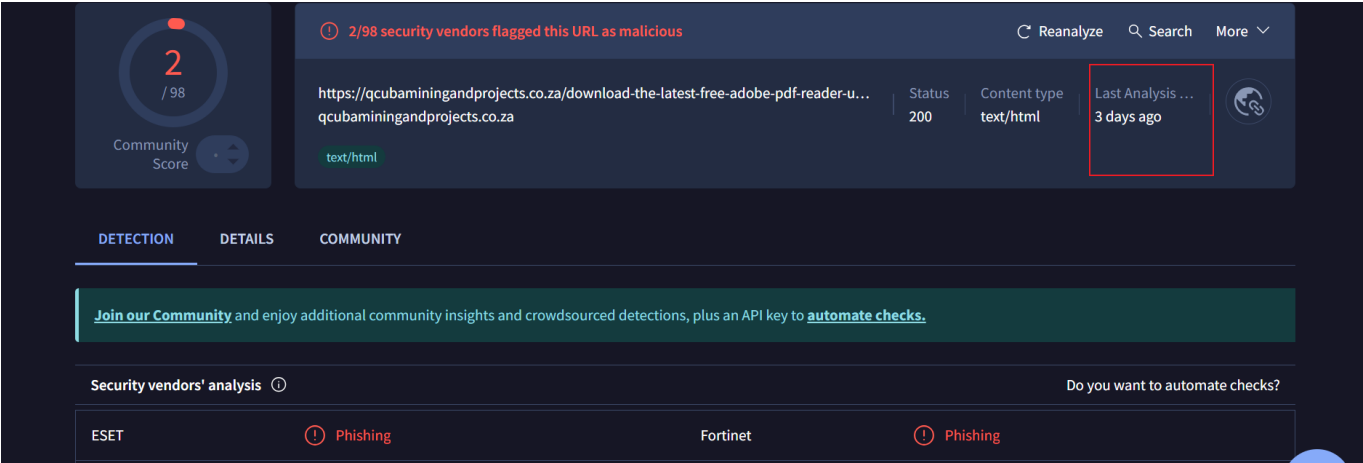
The link was tested using VirusTotal.

12. Malicious Detection



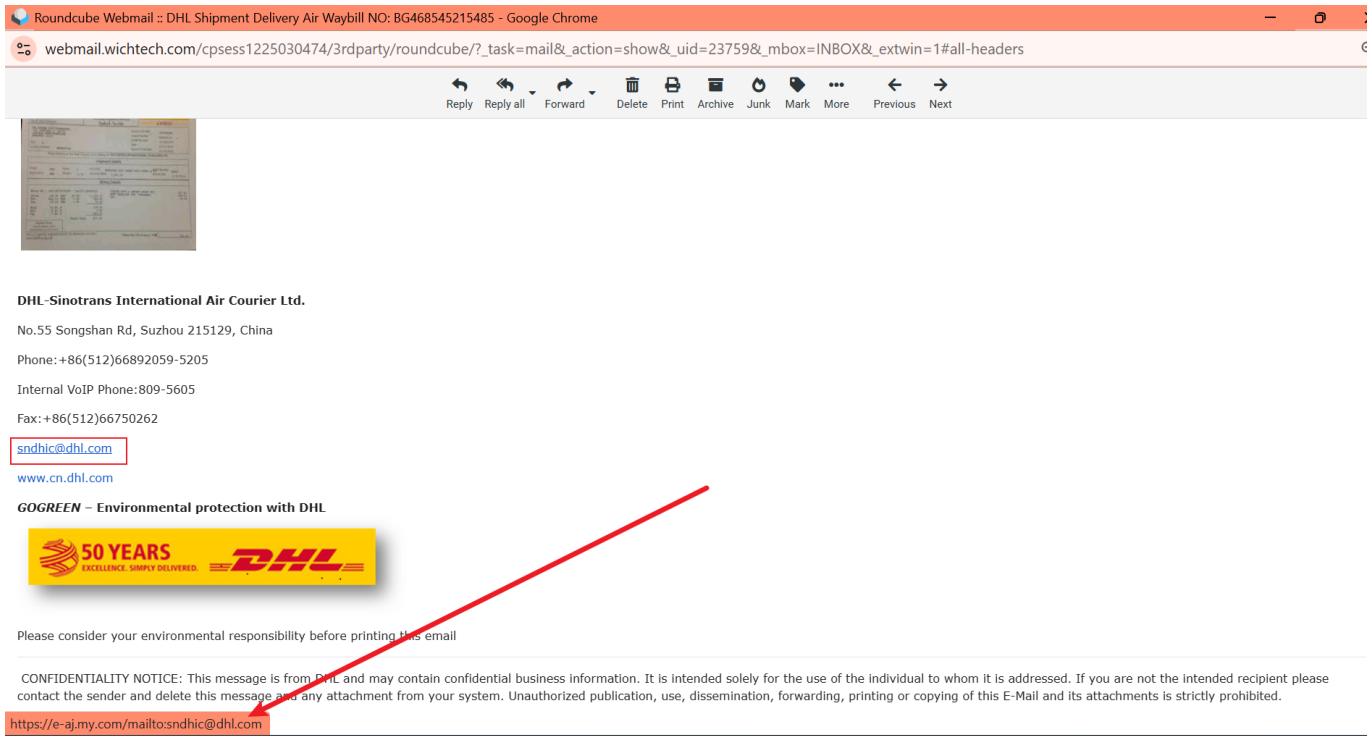
Two security vendors flagged the link as malicious.

13. Prior Phishing Confirmation



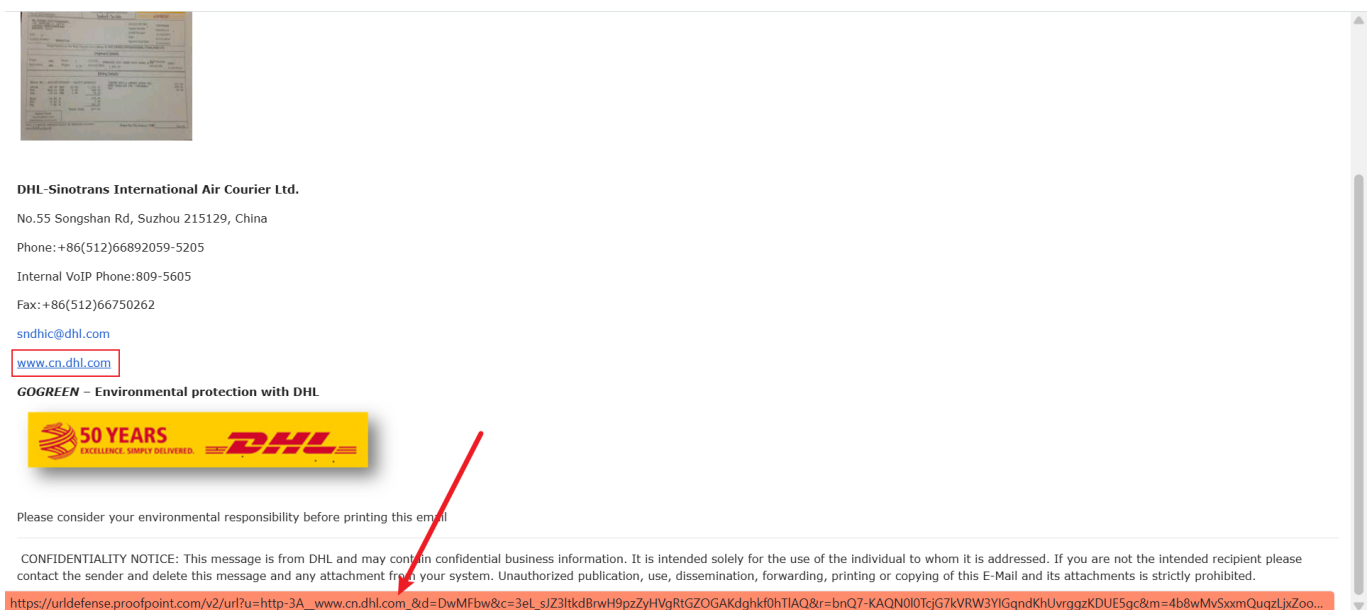
Another user previously identified the link as phishing.

14. Additional Malicious Link Found



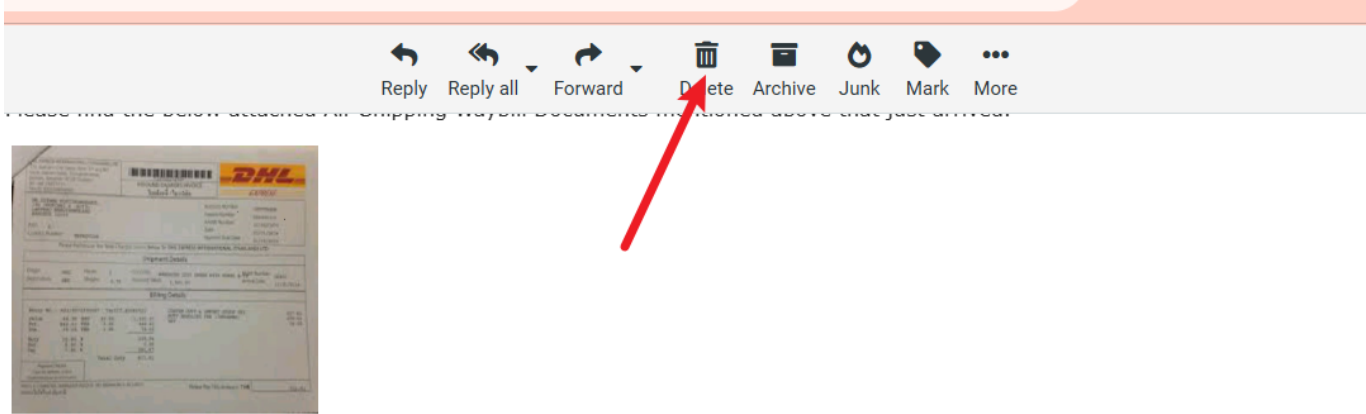
The contact email in the footer was also linked to a known malicious domain.

15. Fake DHL Link Masking



A real DHL link was included to create trust, while malicious links were hidden elsewhere.

16. Final Action



DHL-Sinotrans International Air Courier Ltd.

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sndhic@dhl.com

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The email was deleted or escalated depending on security policy.

Conclusion

This email was confirmed to be a phishing attempt. Key indicators included:

- Sender domain mismatch
- SPF authentication failure
- Suspicious routing path
- Malicious link flagged by VirusTotal

Recommendation

- Do not click suspicious links.
- Always verify sender information.
- Report similar phishing attempts to IT/security.

Below are common terms used in the report explained in plain language:

Term	Simple Explanation
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Term	Simple Explanation
Phishing	A scam where attackers pretend to be someone trustworthy (like a bank or DHL) to trick you into clicking a link or giving information.
Spoofing	When a scammer fakes an email address or name to make it look real.
Email Header	The hidden technical details of an email that show where it came from and the servers it passed through.
Domain	The website or email company's name, e.g., dhl.com . Fake emails often use similar-looking domains.
SPF (Sender Policy Framework)	A security check that verifies if an email is allowed to be sent from a domain. If it fails , the sender is likely fake.
DKIM (DomainKeys Identified Mail)	Another security check that verifies if an email was modified or faked during transit.
Malicious Link	A link designed to cause harm — like stealing passwords, downloading viruses, or redirecting to fake websites.
VirusTotal	A website that checks whether a link or file is safe by comparing it to many security databases.
Urgency-Based Attack	A scam technique where the attacker tries to make you panic and act quickly (like “urgent shipment notice”).
Social Engineering	The psychological manipulation of people to trick them into unsafe decisions.